

Ex. 1

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```
print(uuid::UUIDgenerate())
```

```
## [1] "6f2b31e4-f6aa-4f7b-8f43-6f96bd583427"
```

q1

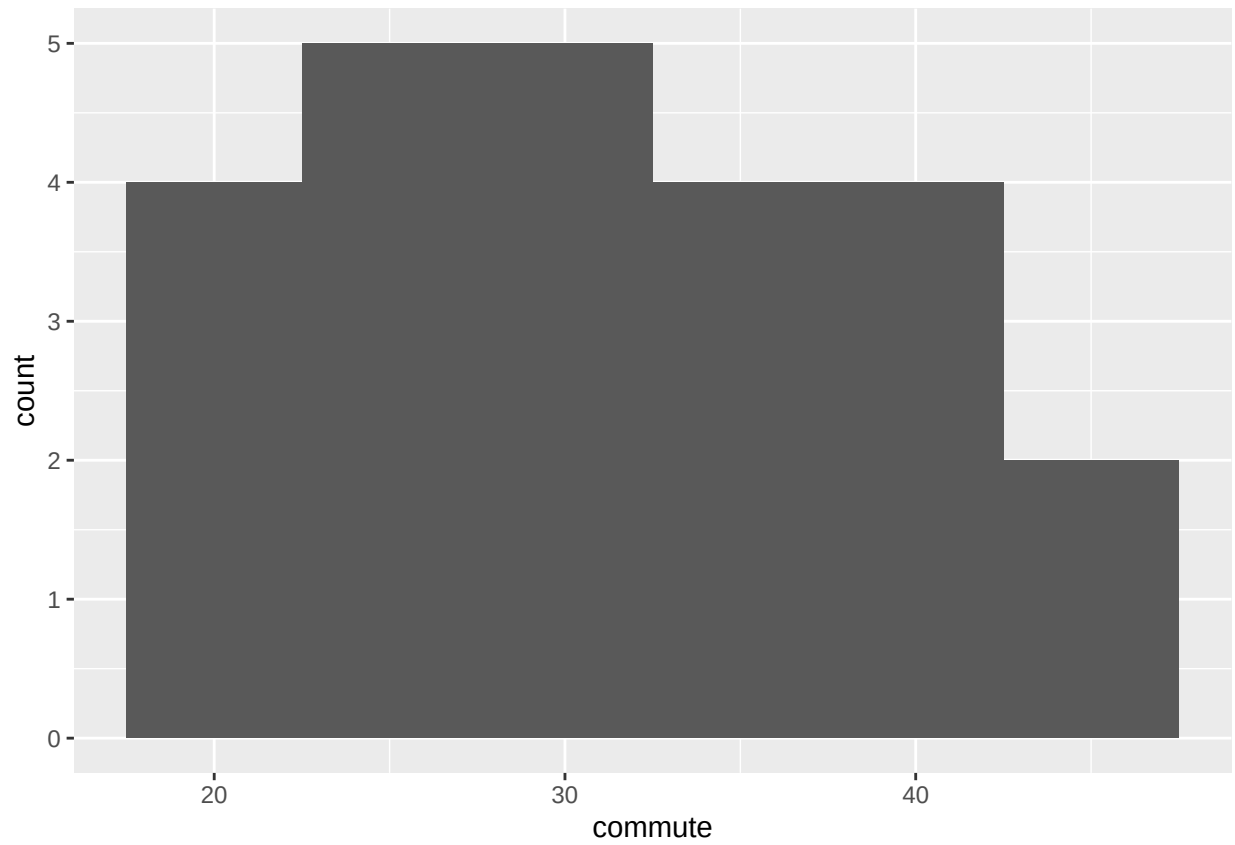
```
library(ggplot2)
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)
# Divide into 5 classes
a <- cut (commute, 5)
# Frequency
freq <- table(a)
# Cumulative frequency
cum_freq <- cumsum(freq)
# Upper class boundaries
upper_boundaries <- c(23.4, 28.8, 34.2, 39.6, 45)
# Create data frame
ogive_data <- data.frame(
  upper_boundary = upper_boundaries,
  cumulative_frequency = cum_freq
)

# q1          freq
freq
```

```
## a
## (18,23.4] (23.4,28.8] (28.8,34.2] (34.2,39.6] (39.6,45]
##          5          5          5          5          4
```

q2

```
# Draw histogram
ggplot(data.frame(commute), aes(x = commute)) +
  geom_histogram(binwidth = 5)
```



```
# Draw ogive
ggplot(ogive_data, aes(x = upper_boundary,
                       y = cumulative_frequency)) +
  geom_line() +
  geom_point()
```

